

From Face Perception to Social Categorization: Multidimensional Scaling Approach to Ethnic Outgroup Homogeneity in Childhood

Luca Cussigh, Luciana Carraro, Luigi Castelli



UNIVERSITÀ
DI PADOVA



"In 2023, Member States issued more than 3.7 million residence permits to non-EU citizens from more than 170 countries."

https://www.europarl.europa.eu/doceo/document/E-10-2025-000294-ASW_EN.html

"The study on fiscal impact of migration is a Science for Policy report by the Joint Research Centre, which provides evidence-based scientific support to the European policymaking process. [...]

The study also finds that removing obstacles to the full labour market integration of migrants can produce considerable positive fiscal benefits for hosting countries in the future."

https://www.europarl.europa.eu/doceo/document/E-10-2025-002052-ASW_EN.html

Categorization's effects

- Social categorization changes how people perceive, evaluate, remember, and treat one another (Amodio & Cikara, 2021; Levy et al., 2023; Rhodes & Baron, 2019)

Why is social categorization a relevant phenomenon in childhood?

- Early-emerging ingroup preferences documented in infants (e.g., Kelly et al., 2009; Woo et al., 2020)
- Visual specialization for ingroup faces (Quinn et al., 2016)

The Outgroup Homogeneity Effect (OHE; Quattrone & Jones, 1981)

- Outgroup faces appear more homogeneous to each other than ingroup faces

The Outgroup Homogeneity Effect (with Fantastic Mr.Fox)

In-group



Outgroup Beans



Outgroup Boggins



Outgroup Bunce



Minority-groups homogeneity effect (Tepper & Gilovich, 2025)

- In a multigroup scenario
- Majority members see different minority outgroups as MORE similar to each other

Minority-groups homogeneity effect (with Fantastic Mr.Fox)



The Multigroup Context

In children, the situation is slightly different:

- No formal position consolidated
- Multigroup paradigms are not employed, even though they are more ecologically valid
- **Gap:** We do not know how children categorize faces in a multigroup environment

Research Question:

- How is the multiethnic perceptual space structured in childhood? Does it support homogeneity theories?
- Does ethnic categorization change across development?

Pre-registered studies: Method & Experimental Design

Sample:

- Study 1: N=249 (moderated, controlled lab setting)
- Study 2: N=101 (unmoderated, online setting)
- Age range: 3 to 8 years

Multidimensional scaling (MDS)

Expanding the Scope

- 4-group context: Italian ingroup + 3 distinct ethnic outgroups
- Relational focus: Facial dyads rather than isolated face evaluation
- Continuous similarity judgments (0-3 scale) instead of binary choices

Face Similarity Task Protocol:

- Incomplete block design
- 66 random dyads presented sequentially

The Face Similarity Task

→	→	→
		Quanto si somigliano? Molto Diversi Un po' Diversi Un po' Simili Molto Simili
800ms	1000ms	//

MDS

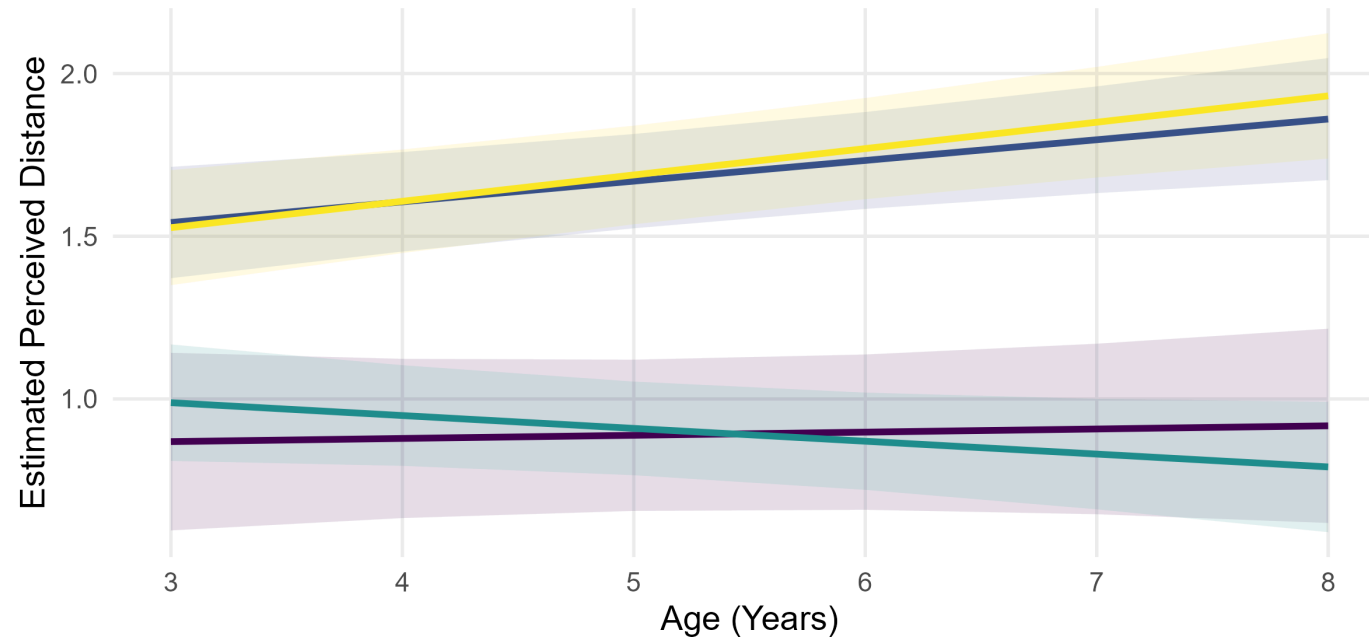
3D MDS visualization

- Children spontaneously visually categorize stimuli in ethnic groups

LMM - Developmental Trajectories

Study 1 (In-Person): Macro Developmental Trajectories

Global Category x Age Interaction



Group Relationship:

- In-In (Italians)
- In-Out (Italian vs Minority)
- Out-Out (Same Minority)
- Out1-Out2 (Different Minorities)

- Perceptual expertise in discriminating different social groups increases with age

- Perceptual expertise in discriminating different social groups increases with age
- Intergroup distinctions become more salient

Questions? 🙌

PRESENTATION AVAILABLE AT

